SPEC SHEET · PRELIMINARY

Crescent City — V2.0

An ESP32-S3 Wi-Fi and Bluetooth LE development board with a built-in 0.96" OLED, USB-C, and protected single-cell battery charging. Breadboard-friendly, fully documented, and running out of the box.

`Board V2.0 · EE2026003B` `Status Preliminary` `Updated October 27, 2026`

What you get

- ESP32-S3, dual-core 240 MHz, 8 MB flash, Wi-Fi 4 and Bluetooth 5 LE
- Complete battery power: USB-C charging, cell protection, automatic USB/battery switching and battery monitoring
- Ships with an open-source Weather Clock; Arduino and ESPHome / Home Assistant ready
- 0.96" 128×64 OLED built in — no wiring, working out of the box
- Qwiic / STEMMA QT connector plus 12 GPIO on breadboard-friendly headers
- Published schematic, 3D model and a 122-test validation report

Specifications

`measured V2.0` `measured V1.0` bench measurement on the V2.0 or V1.0 boards (see the DVT report). `V2.0 target` design value, still to be confirmed in V2.0 testing (October–December 2026). All other values are from the design.

PROCESSOR & WIRELESS

Module	ESP32-S3-WROOM-1-N8 with on-module PCB antenna
Processor	Dual-core Xtensa LX7, up to 240 MHz
Memory	512 KB SRAM · 8 MB flash · no PSRAM
Wi-Fi	802.11 b/g/n (Wi-Fi 4), 2.4 GHz
Bluetooth	Bluetooth 5 (LE)

USB & PROGRAMMING

Connector	USB-C, USB 2.0 full speed (native ESP32-S3 USB)
Used for	Power, charging, programming, serial console and debugging. No driver needed on Windows 10/11, macOS or Linux.
Compatibility	Works with USB-C chargers and C-to-C cables; ESD protection on the data lines
Buttons	BOOT (GPIO0, also usable as a user button) and EN (reset)

DISPLAY

Type	0.96" white monochrome OLED, 128 × 64 pixels
Controller	SSD1306, 4-wire SPI on dedicated pins (not shared with the headers)
Display pins	CS GPIO10 · DC GPIO9 · SCLK GPIO12 · MOSI GPIO11 · RESET GPIO14

POWER & BATTERY

Power input	5 V from USB-C, or the 5V_IN header pin (input only, with a 0.5 A resettable fuse and a blocking diode)
Battery	Single-cell 3.7 V Li-ion / LiPo, JST-PH 2.0 mm connector, polarity marked. 600 mAh or larger recommended.
Charging	303 mA constant current `measured V2.0` to 4.20 V, then tapers and stops at ≈8% of the charge current (23–25 mA) `measured V2.0` V1.0: 201 mA, 4.19 V, stopped at 7.6% `measured V1.0`
Charge time	About 5.4 h per 1000 mAh from empty on the test bench `measured V2.0` 1,888 mAh in 10.3 h. The bench wiring adds resistance, so a cell plugged straight into the board charges a little faster. V1.0 on the bench: about 6.7 h per 1000 mAh `measured V1.0`
Battery protection	Over-charge, over-discharge (cuts off at ≈2.4 V; 2.43 V `measured V2.0`), over-current / short circuit, reverse battery polarity
Power switching	Automatic: runs from USB when plugged in (battery isolated), switches to the battery when USB is removed `measured V2.0`
3.3 V regulator	Low-dropout regulator, 600 mA
Monitoring	Battery voltage on GPIO4 and USB voltage on GPIO2, each through a ½ divider (ADC1)
Low-battery shutdown	The Weather Clock warns at 3.45 V and shuts down cleanly at 3.30 V, before the 3.3 V rail sags, then sleeps until USB is plugged in `measured V2.0` ESPHome configuration: same thresholds `V2.0 target`

POWER CONSUMPTION (TYPICAL, 3.8 V BATTERY)

Weather Clock	≈41 mA — about 24 hours per 1000 mAh `measured V2.0` Firmware 2.1: the Wi-Fi radio is off between weather updates on battery. Earlier firmware with Wi-Fi always on: ≈70 mA, about 14 hours per 1000 mAh `measured V1.0`
Wi-Fi transmit peaks	Up to ≈400 mA for microseconds `measured V1.0`
Deep sleep	≈0.8 mA `measured V2.0` Most of it is the standby current of the always-on LEDs. V1.0: 0.88 mA `measured V1.0`
After protection cutoff	≈1.2 µA `measured V2.0`

LEDS

RGB status LED	Addressable SK6812 on GPIO48
CHG	Yellow, on while charging
SYS ON	Green, on when the 3.3 V rail is up

SOFTWARE

Ships with	Weather Clock: local time and weather over Wi-Fi, with Wi-Fi setup from your phone. Open source (MIT license).
Arduino	“esp32 by Espressif Systems” board package, board “ESP32S3 Dev Module”
ESPHome / Home Assistant	Ready-made configuration: clock and status pages, battery level, Qwiic sensor, RGB LED, updates over Wi-Fi
Other	ESP-IDF, MicroPython and CircuitPython should work with their generic ESP32-S3 builds (not yet tested by us)

I/O & EXPANSION

Header pins	2 × 12, 0.1" (2.54 mm) pitch, rows 0.9" (22.86 mm) apart — fits a standard breadboard
GPIO on headers	12: GPIO 1, 5, 6, 7, 13, 15, 16, 21, 35, 36, 37, 39
Analog inputs	GPIO 1, 5, 6, 7 (ADC1, usable with Wi-Fi on); GPIO 13, 15, 16 (ADC2)
Serial	UART0 on the header (pins marked TX and RX), 3.3 V logic
Qwiic / STEMMA QT	JST-SH 4-pin, I²C on GPIO8 (SDA) and GPIO18 (SCL), 3.3 V, 4.7 kΩ pull-ups
Power pins	3V3 ×3 · GND ×6 · 5V_IN ×1 (input only)
Logic level	3.3 V. GPIO are not 5 V tolerant.

MECHANICAL

Board size	70.1 × 25.4 mm (2.76" × 1.00"); ≈76.4 mm including the antenna overhang
Height	13 mm overall, with the display fitted
Weight	14 g
Mounting holes	None — mount by the header pins or in a breadboard
PCB	4-layer
3D model	STEP file for enclosure design

ENVIRONMENT & COMPLIANCE

Operating temperature	−40 to +85 °C (ESP32-S3 module rating). With a battery fitted, stay within the battery’s own limits.
Charging temperature	0 to 45 °C (battery limit; the charger has no temperature sensor)
Certification	FCC / ISED testing is planned before shipping. The ESP32-S3 module itself is pre-certified.

Header pins

PIN	J4 (BOTTOM ROW)	J5 (TOP ROW)
1	GND	GND
2	3V3	3V3
3	GPIO5	GPIO1
4	GPIO6	GPIO13
5	GPIO7	GPIO39
6	GPIO15	GND
7	GPIO16	GPIO37
8	GND	GPIO36
9	TX (UART0)	GPIO35
10	RX (UART0)	GPIO21
11	GND	GND
12	3V3	5V_IN (input only)

Qwiic connector: GND, 3V3, SDA (GPIO8), SCL (GPIO18). Pin names are printed on both sides of the board. The full diagram with every alternate function is in the pinout PDF.